

Summary Plan: PKS1740-517

ALMA Thesis Planner (automated pipeline)

0.1 Idea

Title: Spatially Resolving CO(2–1) Absorption Against the Double Radio Core of PKS1740–517: One Component or Two?

Idea: Using the publicly available, pipeline-calibrated Band 4 long-baseline (0.034") ALMA data and QA2 image/cube products from project 2018.1.00998.S, characterize the mm continuum morphology of PKS1740-517 and determine whether the CO(2–1) absorption line (rest 230.538 GHz; falls at an observed frequency of ≈ 160.10 GHz in the 159.0–160.9 GHz spectral window, assuming the quoted host redshift $z = 0.44$ — treat this as provisional and verify against Splatalogue before any line identification or imaging) is seen against one or both components of the continuum source. The student would: (1) download the calibrated measurement set and QA2 continuum image for the single member OUS, (2) use the two continuum-only windows (145.0–149.0 GHz observed) to build a deep, high-fidelity continuum map at 0.034" resolution and identify whether the background source resolves into one or two compact components (testing the "double core" merger scenario suggested in the proposal abstract), (3) extract absorption spectra at each identified continuum peak location from the 159.0–160.9 GHz cube using CASA, and (4) compare line depth, centroid velocity, and width between sightlines to determine whether the absorbing gas has a single, spatially uniform kinematic signature (consistent with a compact circumnuclear disk/torus) or differs between components (consistent with extended, merger-disturbed gas). This is a focused measurement-and-comparison project built entirely on existing pipeline products plus modest CASA re-imaging/spectral extraction — not new calibration or observation — and is scoped to answer a single, well-defined geometric question (one absorbing screen or two) rather than building a full kinematic model of the system.

0.2 Methods

1. Data retrieval and inventory

Have the student log into the ALMA Science Archive and pull down the full data package for member OUS 'uid://A001/X133d/X18e9' (project 2018.1.00998.S, target PKS1740-517). This includes the pipeline-calibrated measurement set, the QA2 weblog, and the delivered QA2 image/cube products. Have them read the QA2 weblog first — it records the exact pipeline/CASA version used for calibration, the final flagging summary, and the achieved sensitivities, which the student will need later to sanity-check their own re-imaging. Since long-baseline Band 4 data are involved, have the student confirm from the weblog whether any self-calibration or phase referencing issues were flagged, since these affect how aggressively they can re-image at full resolution.

2. Software environment

Install the CASA version matching (or the closest supported version to) the one recorded in the QA2

weblog, so that pipeline calibration tables and MS structure remain compatible with re-imaging commands ('`tclean`', '`imfit`', '`uvmodelfit`', '`specfit`', '`exportfits`', etc.). Keep a lab notebook of every CASA task and parameter set used, since imaging at $0.034''$ resolution is sensitive to weighting choices and small parameter changes can shift apparent source structure.

3. Continuum imaging of the line-free windows

Use only the two spectral windows identified as continuum-only at this redshift (observed 145.0–147.0 GHz and 147.0–149.0 GHz; note these correspond to rest-frame ≈ 208.8 –214.6 GHz under the assumed $z = 0.44$, and this line-free identification should also be spot-checked against Splatalogue before being treated as final). Flag any residual line contamination if found. Run '`tclean`' in '`specmode='mfs'`' on the combined line-free channels to produce the deepest possible continuum map at native long-baseline resolution, targeting the archive-quoted continuum sensitivity of 0.0119 mJy/bm as a benchmark for the student's own image RMS. Experiment with Briggs weighting (a robust parameter scan, e.g., `robust = 0, 0.5, 1`) to balance resolution against sidelobe/noise behavior at this extreme baseline length, and document how the apparent source morphology changes across the scan.

4. Continuum source characterization

On the best continuum image, have the student search for compact structure at the phase center and identify all significant peaks above, e.g., 5 – 7σ . For each candidate component, fit a point-source or Gaussian model directly in the image plane with '`imfit`', and cross-check with a uv-plane fit ('`uvmodelfit`') using one- and two-component point-source models, comparing fit residuals and reduced chi-squared to determine whether a single-component or double-component model is statistically preferred. Record best-fit positions (RA/Dec), peak flux densities, and formal position uncertainties (beam-size/SNR-scaled) for each component — these positions are the sightlines used in the next step. This directly tests the "double radio core" scenario motivating the project.

5. Verification of the CO(2–1) line identification

Before touching the line-bearing window, have the student independently verify the line identification in Splatalogue: look up CO(2–1) rest frequency (230.538 GHz) and confirm that, under the quoted host redshift $z = 0.44$ (observed = rest/1.44), the expected observed frequency (≈ 160.10 GHz, provisional) falls inside the delivered 159.0–160.9 GHz spectral window. Have the student note explicitly in their methods write-up that this identification and frequency are provisional/assumed pending this check, and flag (but do not attempt to observationally resolve) the CN and HC₃N lines noted as plausible serendipitous contaminants in the neighboring 157.0–159.0 GHz window, so that any spectral features near the CO(2–1) window are correctly attributed.

6. Spectral cube imaging

Image the 159.0–160.9 GHz spectral window as a full cube with '`tclean`' in '`specmode='cube'`', using the native/delivered velocity resolution (≈ 1.84 km/s channels) or a modest binning if SNR requires it, and using the same weighting scheme validated in step 3 for consistency with the continuum astrometry. Continuum-subtract the cube ('`uvcontsub`' or image-plane subtraction using nearby line-free channels within the same window, if any) so that absorption is measured relative to the correct local continuum level rather than the broadband continuum from step 3.

7. Absorption spectrum extraction at each continuum peak

At the position of each continuum component identified in step 4, extract a spectrum from the continuum-subtracted cube (single-pixel or small-aperture extraction matched to the synthe-

sized beam) using CASA’s ‘imval’/‘specflux’-type tools or a Python/‘spectral-cube’ post-processing script. Normalize each extracted spectrum by the local continuum flux density at that position (from the step-3/step-4 continuum fit) to produce an optical-depth-like absorption profile (line depth relative to continuum) for each sightline.

8. Line profile fitting and comparison

Fit each extracted absorption profile with a Gaussian (or multi-Gaussian, if profiles are clearly multi-component) using ‘specfit’ or a Python least-squares routine, deriving for each sightline: peak absorption depth (or apparent optical depth), centroid velocity (relative to the assumed systemic redshift $z = 0.44$), and FWHM line width, each with formal fit uncertainties propagated from the cube RMS. Tabulate these parameters side by side for each continuum component.

9. Geometric interpretation

Compare the fitted parameters across sightlines using their formal uncertainties: if centroid velocities, widths, and depths agree within errors between components, treat this as evidence for a single, spatially uniform absorbing screen (consistent with a compact circumnuclear disk/torus); if they differ significantly, treat this as evidence that the absorption differs between components (consistent with extended, merger-disturbed gas straddling a double core). Where only one continuum component is robustly detected, adjust the analysis to report a single-sightline absorption profile and state that the two-component test could not be performed. Present the final continuum map, the extracted/fitted absorption spectra per sightline, and the tabulated comparison of centroid velocity, depth, and width as the core results answering the one-screen-versus-two-screen question.